

Security Scan Report

Target: C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberSceurity
Project - Copy\security-signal-engine\test_vulnerable_app

Scan Date: July 24, 2026 at 05:50 UTC

Scan ID: 81a5b376-019f-4378-964e-2621a1e97895

0 CRITICAL

2 HIGH

3 MEDIUM

4 LOW

Executive Summary

Security scan of 'C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberSceurity Project - Copy\security-signal-engine\test_vulnerable_app' identified 39 raw findings, reduced to 9 unique issues after deduplication. ■ 2 HIGH-severity issue(s) should be prioritized for remediation within the next 48 hours. 3 MEDIUM-priority issue(s) should be addressed in the current development sprint. 4 LOW-priority issue(s) have been logged for backlog tracking.

Scan Statistics

Metric	Value
Total Raw Findings	39
After Deduplication	9
Critical Findings	0
High Findings	2
Medium Findings	3
Low Findings	4
Scan Duration	11.2s

Detailed Findings

[HIGH] #1: Ssrf in C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\test_vulnerable_app\vulnerable_server.py:310

Risk Score	0.6174
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\te
Vulnerability Type	SSRF
CWE ID	CWE-918: Server-Side Request Forgery (SSRF)
Detected By	semgrep

Explanation
Server-Side Request Forgery (SSRF) lets attackers make the server send HTTP requests to arbitrary URLs, including internal services. Found in vulnerable_server.py. Data from request object is passed to a new server-side request.

Code or HTTP Excerpt

```
308 | def fetch_url():
309 |     """■ SSRF: Server fetches any URL provided by the user."""
>>> 310 |     url = request.args.get("url", "")
311 |     content = ""
312 |     <h2>URL Fetcher</h2>
```

Business Impact
An attacker could access user accounts and personal data from within the server's network, potentially reading cloud metadata, internal APIs, or admin panels.

Fix Recommendation
Best Fix: Whitelist allowed domains for outbound requests. Block private IP ranges. Validate and resolve URLs before making requests.

[MEDIUM] #2: Open Redirect in C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\test_vulnerable_app\vulnerable_server.py:298

Risk Score	0.4508
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\te
Vulnerability Type	OPEN_REDIRECT
CWE ID	CWE-601: URL Redirection to Untrusted Site ('Open Redirect')
Detected By	semgrep

Explanation
Open redirect allows attackers to redirect users from your trusted domain to a malicious site via a URL parameter. Found in vulnerable_server.py. Data from request is passed to redirect().

Code or HTTP Excerpt

```
296 | def open_redirect():
297 |     """■ Open Redirect: Redirects to any URL provided by the user."""
>>> 298 |     url = request.args.get("url", "/")
299 |     # ■ VULNERABLE: No validation of redirect target
300 |     return redirect(url)
```

Business Impact

Attackers could send users convincing phishing links via user accounts and personal data that appear to originate from your trusted domain.

Fix Recommendation

Best Fix: Validate redirect URLs against a whitelist of allowed domains. Use relative paths for internal redirects. Reject external URLs.

[MEDIUM] #3: Unknown in C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\test_vulnerable_app\vulnerable_server.py:463

Risk Score	0.4508
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\te
Vulnerability Type	UNKNOWN
CWE ID	CWE-611: Improper Restriction of XML External Entity Reference
Detected By	semgrep

Explanation

A security vulnerability was detected in your code. Found in vulnerable_server.py. The Python documentation recommends using `defusedxml` instead of `xml` because the native Python `xml` library is vulnerable to XML External Entity (XXE) attacks.

Code or HTTP Excerpt

```
461 |         return render_template_string(LAYOUT, title="XML API", content=content)
462 |
>>> 463 |     import xml.etree.ElementTree as ET
464 |     try:
465 |         # ■ VULNERABLE: Parsing XML without disabling external entities
```

Business Impact

This vulnerability could be exploited to compromise user accounts and personal data.

Fix Recommendation

Best Fix: Review the flagged code and apply security best practices. Consult the OWASP guidelines for the relevant vulnerability type.

[HIGH] #4: Sql Injection in C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\test_vulnerable_app\vulnerable_server.py:160

Risk Score	0.4324
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\te

Vulnerability Type	SQL_INJECTION
CWE ID	CWE-89: Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection')
Detected By	semgrep

Explanation
SQL Injection occurs when user-controlled input is directly interpolated into SQL queries, allowing an attacker to alter query logic. Found in vulnerable_server.py. User-controlled data from a request is passed to 'execute()'.

Code or HTTP Excerpt

```
158 | def users():
159 |     """■ SQL Injection: User input directly concatenated into SQL."""
>>> 160 |     user_id = request.args.get("id", "")
161 |     content = "<h2>User Lookup</h2>"
162 |     content += '<form action="/users" method="get">'
```

Business Impact
An attacker could read, modify, or delete user accounts and personal data, potentially exfiltrating the entire database including credentials and PII.

Fix Recommendation
Best Fix: Use Django ORM queries instead of raw SQL: User.objects.filter(id=user_id). If raw SQL is required, pass parameters as a list: cursor.execute('...WHERE id=%s', [user_id]).

[MEDIUM] #5: Unknown in C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\test_vulnerable_app\vulnerable_server.py:149

Risk Score	0.4230
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\te
Vulnerability Type	UNKNOWN
CWE ID	CWE-614: Sensitive Cookie in HTTPS Session Without 'Secure' Attribute
Detected By	semgrep

Explanation
A security vulnerability was detected in your code. Found in vulnerable_server.py. Found a Flask cookie with insecurely configured properties.

Code or HTTP Excerpt

```
147 |     # ■ Insecure cookie
148 |     resp.set_cookie("session_id", "abc123", httponly=False, secure=False, samesite=None)
>>> 149 |     resp.set_cookie("user_token", "eyJhbGciOiJIUzI1NiJ9.test", httponly=False)
150 |     return resp
151 |
```

Business Impact
This vulnerability could be exploited to compromise user session data.

Fix Recommendation
Best Fix: Review the flagged code and apply security best practices. Consult the OWASP guidelines for the relevant vulnerability type.

[LOW] #6: Unknown in C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\test_vulnerable_app\vulnerable_server.py:171

Risk Score	0.3920
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\te
Vulnerability Type	UNKNOWN
CWE ID	CWE-704: Incorrect Type Conversion or Cast
Detected By	semgrep

Explanation

A security vulnerability was detected in your code. Found in vulnerable_server.py. Detected user input used to manually construct a SQL string.

Code or HTTP Excerpt

```
169 |         try:
170 |             # ■ VULNERABLE: Direct string concatenation in SQL
>>> 171 |             query = f"SELECT * FROM users WHERE id = {user_id}"
      172 |             cursor.execute(query)
      173 |             rows = cursor.fetchall()
```

Business Impact

This vulnerability could be exploited to compromise the database.

Fix Recommendation

Best Fix: Review the flagged code and apply security best practices. Consult the OWASP guidelines for the relevant vulnerability type.

[LOW] #7: Xss in C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\test_vulnerable_app\vulnerable_server.py:163

Risk Score	0.3478
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\te
Vulnerability Type	XSS
CWE ID	CWE-79: Improper Neutralization of Input During Web Page Generation ('Cross-site Scripting')
Detected By	semgrep

Explanation

Cross-Site Scripting (XSS) allows attackers to inject malicious scripts into web pages viewed by other users. Found in vulnerable_server.py. Detected user input flowing into a manually constructed HTML string.

Code or HTTP Excerpt

```
161 |         content = "<h2>User Lookup</h2>"
162 |         content += '<form action="/users" method="get">'
>>> 163 |         content += '<input name="id" placeholder="Enter user ID" value="' + user_id + '">'
```

```
164 |         content += '<button type="submit">Search</button></form>'
165 |
```

Business Impact

An attacker could steal session tokens from user accounts and personal data, redirect users to phishing pages, or perform actions as the victim.

Fix Recommendation

Best Fix: Django templates auto-escape by default. Avoid using `|safe` or `mark_safe()` on user-controlled content.

[LOW] #8: Unknown in C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\test_vulnerable_app\vulnerable_server.py:124

Risk Score	0.3384
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\te
Vulnerability Type	UNKNOWN
CWE ID	CWE-96: Improper Neutralization of Directives in Statically Saved Code ('Static Code Injection')
Detected By	semgrep

Explanation

A security vulnerability was detected in your code. Found in `vulnerable_server.py`. Found a template created with string formatting.

Code or HTTP Excerpt

```
122 |         # ■ Missing security headers (X-Frame-Options, CSP, etc.)
123 |         # ■ Server version disclosure
>>> 124 |         resp = make_response(render_template_string(LAYOUT, title="Home", content="")
125 |                               <p>Welcome to VulnApp – an intentionally vulnerable application.</p>
126 |                               <h2>Navigation</h2>
```

Business Impact

This vulnerability could be exploited to compromise user accounts and personal data.

Fix Recommendation

Best Fix: Review the flagged code and apply security best practices. Consult the OWASP guidelines for the relevant vulnerability type.

[LOW] #9: Unknown in C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\test_vulnerable_app\vulnerable_server.py:321

Risk Score	0.3384
Affected Location	C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\te
Vulnerability Type	UNKNOWN
CWE ID	CWE-939: Improper Authorization in Handler for Custom URL Scheme

Detected By

semgrep

Explanation

A security vulnerability was detected in your code. Found in vulnerable_server.py. Detected a dynamic value being used with urllib.

Code or HTTP Excerpt

```
319 |         try:
320 |             # ■ VULNERABLE: No URL validation, allows internal network access
>>> 321 |             resp = urllib.request.urlopen(url, timeout=5)
322 |             body = resp.read(4096).decode("utf-8", errors="replace")
323 |             content += f"<h3>Response from {url}</h3>"
```

Business Impact

This vulnerability could be exploited to compromise user accounts and personal data.

Fix Recommendation

Best Fix: Review the flagged code and apply security best practices. Consult the OWASP guidelines for the relevant vulnerability type.

Appendix

Deduplication Statistics

Total raw findings: 39 | After deduplication: 9 | Reduction: 77%

Scan Metadata

Scan ID	81a5b376-019f-4378-964e-2621a1e97895
Target	C:\Users\Dhruv\Desktop\College Stuff\Projects\CyberScurity Project - Copy\security-signal-engine\te
Timestamp	2026-07-24T05:50:55.637125+00:00
Duration	11.2s
Scanners Used	semgrep
Scanner Versions	semgrep=1.159.0

Raw Findings Data (JSON)

```
[
  {
    "id": "5fff038e-f5a1-4178-8a51-9d786eb5acb8",
    "title": "Ssrf in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\secu",
    "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sig",
    "line_start": 310,
    "line_end": 326,
    "severity": "HIGH",
    "vuln_type": "SSRF",
    "description": "Data from request object is passed to a new server-side request. This could lead to a s",
    "rule_id": "python.django.security.injection.ssrfs.ssrfinjectionurllib.ssrfinjectionurllib",
    "confidence": 0.9,
    "cwe_id": "CWE-918: Server-Side Request Forgery (SSRF)",
    "cvss_score": null,
    "code_snippet": "
        308 | def fetch_url():\n
        309 |         \"\"\"u274c SSRF: Server fetches any URL p
    "sources": [
      "semgrep"
    ],
    "risk_score": 0.6174,
    "risk_category": "HIGH",
    "category_source": "score",
    "category_reason": "score-threshold",
    "llm_explanation": "Server-Side Request Forgery (SSRF) lets attackers make the server send HTTP requests",
    "fix_suggestion": "",
    "business_impact": "An attacker could access user accounts and personal data from within the server's ne",
    "raw": {
      "check_id": "python.django.security.injection.ssrfs.ssrfinjectionurllib.ssrfinjectionurllib",
      "path": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-s",
      "start": {
        "line": 310,
        "col": 5,
        "offset": 13511
      },
      "end": {
        "line": 326,
        "col": 68,
        "offset": 14219
      },
      "extra": {
```

```

"message": "Data from request object is passed to a new server-side request. This could lead to a s
"metadata": {
  "cwe": [
    "CWE-918: Server-Side Request Forgery (SSRF)"
  ],
  "owasp": [
    "A10:2021 - Server-Side Request Forgery (SSRF)",
    "A01:2025 - Broken Access Control"
  ],
  "references": [
    "https://owasp.org/www-community/attacks/Server_Side_Request_Forgery"
  ],
  "category": "security",
  "technology": [
    "django"
  ],
  "cwe2022-top25": true,
  "cwe2021-top25": true,
  "subcategory": [
    "vuln"
  ],
  "likelihood": "MEDIUM",
  "impact": "HIGH",
  "confidence": "MEDIUM",
  "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
  "vulnerability_class": [
    "Server-Side Request Forgery (SSRF)"
  ],
  "source": "https://semgrep.dev/r/python.django.security.injection.ssrfs.ssrfs-injection-urllib.ssrfs",
  "shortlink": "https://sg.run/6n2B"
},
"severity": "ERROR",
"fingerprint": "requires login",
"lines": "requires login",
"validation_state": "NO_VALIDATOR",
"engine_kind": "OSS"
}
},
{
  "id": "a474ef70-cf6c-44a5-85e3-b0aec19b7778",
  "title": "Open Redirect in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - C
  "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sign
  "line_start": 298,
  "line_end": 300,
  "severity": "HIGH",
  "vuln_type": "OPEN_REDIRECT",
  "description": "Data from request is passed to redirect(). This is an open redirect and could be exploit
  "rule_id": "python.flask.security.open-redirect.open-redirect",
  "confidence": 0.9,
  "cwe_id": "CWE-601: URL Redirection to Untrusted Site ('Open Redirect')",
  "cvss_score": null,
  "code_snippet": "    296 | def open_redirect():\n    297 |         \"\"\"Open Redirect: Redirects t
  "sources": [
    "semgrep"
  ],
  "risk_score": 0.4508,
  "risk_category": "MEDIUM",
  "category_source": "score",
  "category_reason": "score-threshold",
  "llm_explanation": "Open redirect allows attackers to redirect users from your trusted domain to a mali
  "fix_suggestion": "",
  "business_impact": "Attackers could send users convincing phishing links via user accounts and personal
  "raw": {
    "check_id": "python.flask.security.open-redirect.open-redirect",
    "path": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-s
    "start": {
      "line": 298,
      "col": 5,
      "offset": 12886
    }
  },

```

```

"end": {
  "line": 300,
  "col": 25,
  "offset": 13000
},
"extra": {
  "message": "Data from request is passed to redirect(). This is an open redirect and could be exploited",
  "metadata": {
    "cwe": [
      "CWE-601: URL Redirection to Untrusted Site ('Open Redirect')"
    ],
    "owasp": [
      "A01:2021 - Broken Access Control",
      "A01:2025 - Broken Access Control"
    ],
    "references": [
      "https://flask-login.readthedocs.io/en/latest/#login-example",
      "https://cheatsheetseries.owasp.org/cheatsheets/Unvalidated_Redirects_and_Forwards_Cheat_Sheet.html",
      "https://docs.python.org/3/library/urllib.parse.html#url-parsing"
    ],
    "category": "security",
    "technology": [
      "flask"
    ],
    "subcategory": [
      "audit"
    ],
    "likelihood": "LOW",
    "impact": "MEDIUM",
    "confidence": "LOW",
    "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
    "vulnerability_class": [
      "Open Redirect"
    ],
    "source": "https://semgrep.dev/r/python.flask.security.open-redirect.open-redirect",
    "shortlink": "https://sg.run/kXe2"
  },
  "severity": "ERROR",
  "fingerprint": "requires login",
  "lines": "requires login",
  "validation_state": "NO_VALIDATOR",
  "engine_kind": "OSS"
}
},
{
  "id": "73854ab3-afc4-43e9-bc82-97daca5d364b",
  "title": "Unknown in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sign",
  "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sign",
  "line_start": 463,
  "line_end": 463,
  "severity": "HIGH",
  "vuln_type": "UNKNOWN",
  "description": "The Python documentation recommends using `defusedxml` instead of `xml` because the native",
  "rule_id": "python.lang.security.use-defused-xml.use-defused-xml",
  "confidence": 0.9,
  "cwe_id": "CWE-611: Improper Restriction of XML External Entity Reference",
  "cvss_score": null,
  "code_snippet": "461 |         return render_template_string(LAYOUT, title=\"XML API\", content=content)"
  "sources": [
    "semgrep"
  ],
  "risk_score": 0.4508,
  "risk_category": "MEDIUM",
  "category_source": "score",
  "category_reason": "score-threshold",
  "llm_explanation": "A security vulnerability was detected in your code. Found in vulnerable_server.py. The",
  "fix_suggestion": "",
  "business_impact": "This vulnerability could be exploited to compromise user accounts and personal data",
  "raw": {
    "check_id": "python.lang.security.use-defused-xml.use-defused-xml",

```

```

"path": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sig",
"start": {
  "line": 463,
  "col": 5,
  "offset": 20271
},
"end": {
  "line": 463,
  "col": 39,
  "offset": 20305
},
"extra": {
  "message": "The Python documentation recommends using `defusedxml` instead of `xml` because the nat",
  "metadata": {
    "owasp": [
      "A04:2017 - XML External Entities (XXE)",
      "A05:2021 - Security Misconfiguration",
      "A02:2025 - Security Misconfiguration"
    ],
    "cwe": [
      "CWE-611: Improper Restriction of XML External Entity Reference"
    ],
    "references": [
      "https://docs.python.org/3/library/xml.html",
      "https://github.com/tiran/defusedxml",
      "https://owasp.org/www-community/vulnerabilities/XML_External_Entity_(XXE)_Processing"
    ],
    "category": "security",
    "technology": [
      "python"
    ],
    "cwe2022-top25": true,
    "cwe2021-top25": true,
    "subcategory": [
      "audit"
    ],
    "likelihood": "LOW",
    "impact": "MEDIUM",
    "confidence": "LOW",
    "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
    "vulnerability_class": [
      "XML Injection"
    ],
    "source": "https://semgrep.dev/r/python.lang.security.use-defused-xml.use-defused-xml",
    "shortlink": "https://sg.run/kX47"
  },
  "severity": "ERROR",
  "fingerprint": "requires login",
  "lines": "requires login",
  "validation_state": "NO_VALIDATOR",
  "engine_kind": "OSS"
}
},
{
  "id": "a042165f-aeab-4c67-9459-ce49ba75c859",
  "title": "Sql Injection in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - C",
  "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sig",
  "line_start": 160,
  "line_end": 185,
  "severity": "MEDIUM",
  "vuln_type": "SQL_INJECTION",
  "description": "User-controlled data from a request is passed to 'execute()'. This could lead to a SQL i",
  "rule_id": "python.django.security.injection.sql.sql-injection-using-db-cursor-execute.sql-injection-db",
  "confidence": 0.7,
  "cwe_id": "CWE-89: Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection')",
  "cvss_score": null,
  "code_snippet": "      158 | def users():\n      159 |         \"\"\"SQL Injection: User input directly",
  "sources": [
    "semgrep"
  ],

```

```

    "risk_score": 0.4324,
    "risk_category": "HIGH",
    "category_source": "vuln_floor",
    "category_reason": "minimum floor for SQL_INJECTION",
    "llm_explanation": "SQL Injection occurs when user-controlled input is directly interpolated into SQL qu
    "fix_suggestion": "",
    "business_impact": "An attacker could read, modify, or delete user accounts and personal data, potentia
    "raw": {
      "check_id": "python.django.security.injection.sql.sql-injection-using-db-cursor-execute.sql-injection.
      "path": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-s
      "start": {
        "line": 160,
        "col": 5,
        "offset": 6608
      },
      "end": {
        "line": 185,
        "col": 21,
        "offset": 7824
      },
      "extra": {
        "message": "User-controlled data from a request is passed to 'execute()'. This could lead to a SQL :
        "metadata": {
          "cwe": [
            "CWE-89: Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection')"
          ],
          "owasp": [
            "A01:2017 - Injection",
            "A03:2021 - Injection",
            "A05:2025 - Injection"
          ],
          "references": [
            "https://docs.djangoproject.com/en/3.0/topics/security/#sql-injection-protection"
          ],
          "category": "security",
          "technology": [
            "django"
          ],
          "cwe2022-top25": true,
          "cwe2021-top25": true,
          "subcategory": [
            "vuln"
          ],
          "likelihood": "MEDIUM",
          "impact": "HIGH",
          "confidence": "MEDIUM",
          "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
          "vulnerability_class": [
            "SQL Injection"
          ],
          "source": "https://semgrep.dev/r/python.django.security.injection.sql.sql-injection-using-db-curs
          "shortlink": "https://sg.run/qx7y"
        },
        "severity": "WARNING",
        "fingerprint": "requires login",
        "lines": "requires login",
        "validation_state": "NO_VALIDATOR",
        "engine_kind": "OSS"
      }
    }
  },
  {
    "id": "d6d31823-b9ce-45c2-a13d-05b857173f83",
    "title": "Unknown in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\s
    "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sign
    "line_start": 149,
    "line_end": 149,
    "severity": "MEDIUM",
    "vuln_type": "UNKNOWN",
    "description": "Found a Flask cookie with insecurely configured properties. By default the secure, http
    "rule_id": "python.flask.security.audit.secure-set-cookie.secure-set-cookie",

```

```

"confidence": 0.7,
"cwe_id": "CWE-614: Sensitive Cookie in HTTPS Session Without 'Secure' Attribute",
"cvss_score": null,
"code_snippet": "      147 |      # \u274c Insecure cookie\n      148 |      resp.set_cookie(\"session_id\",
\"sources\": [
    \"semgrep\"
],
\"risk_score\": 0.423,
\"risk_category\": \"MEDIUM\",
\"category_source\": \"score\",
\"category_reason\": \"score-threshold\",
\"llm_explanation\": \"A security vulnerability was detected in your code. Found in vulnerable_server.py. F\",
\"fix_suggestion\": \"\",
\"business_impact\": \"This vulnerability could be exploited to compromise user session data.\",
\"raw\": {
    \"check_id\": \"python.flask.security.audit.secure-set-cookie.secure-set-cookie\",
    \"path\": \"C:\\\\Users\\\\Dhruv\\\\Desktop\\\\College Stuff\\\\Projects\\\\CyberScurity Project - Copy\\\\security-s\",
    \"start\": {
        \"line\": 149,
        \"col\": 5,
        \"offset\": 5991
    },
    \"end\": {
        \"line\": 149,
        \"col\": 79,
        \"offset\": 6065
    },
    \"extra\": {
        \"message\": \"Found a Flask cookie with insecurely configured properties. By default the secure, http\",
        \"metadata\": {
            \"cwe\": [
                \"CWE-614: Sensitive Cookie in HTTPS Session Without 'Secure' Attribute\"
            ],
            \"owasp\": [
                \"A05:2021 - Security Misconfiguration\",
                \"A02:2025 - Security Misconfiguration\"
            ],
            \"references\": [
                \"https://flask.palletsprojects.com/en/3.0.x/api/#flask.Response.set_cookie\",
                \"https://flask.palletsprojects.com/en/3.0.x/security/#set-cookie-options\"
            ],
            \"category\": \"security\",
            \"technology\": [
                \"python\",
                \"flask\"
            ],
            \"subcategory\": [
                \"audit\"
            ],
            \"likelihood\": \"LOW\",
            \"impact\": \"LOW\",
            \"confidence\": \"LOW\",
            \"functional-categories\": [
                \"web::search::cookie-config::flask\"
            ],
            \"license\": \"Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license\",
            \"vulnerability_class\": [
                \"Cookie Security\"
            ],
            \"source\": \"https://semgrep.dev/r/python.flask.security.audit.secure-set-cookie.secure-set-cookie\",
            \"shortlink\": \"https://sg.run/gLkZ\"
        },
        \"severity\": \"WARNING\",
        \"fingerprint\": \"requires login\",
        \"lines\": \"requires login\",
        \"validation_state\": \"NO_VALIDATOR\",
        \"engine_kind\": \"OSS\"
    }
}
},
{

```

```

"id": "b1569d0d-460a-4b64-bea1-27ad81633863",
"title": "Unknown in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\",
"file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sign",
"line_start": 171,
"line_end": 171,
"severity": "HIGH",
"vuln_type": "UNKNOWN",
"description": "Detected user input used to manually construct a SQL string. This is usually bad practice",
"rule_id": "python.flask.security.injection.tainted-sql-string.tainted-sql-string",
"confidence": 0.9,
"cwe_id": "CWE-704: Incorrect Type Conversion or Cast",
"cvss_score": null,
"code_snippet": "        169 |                try:\\n            170 |                    # \\u274c VULNERABLE: Direct string concatenation",
"sources": [
    "semgrep"
],
"risk_score": 0.392,
"risk_category": "LOW",
"category_source": "score",
"category_reason": "score-threshold",
"llm_explanation": "A security vulnerability was detected in your code. Found in vulnerable_server.py. LLM explanation: Detected user input used to manually construct a SQL string. This is usually bad practice",
"fix_suggestion": "",
"business_impact": "This vulnerability could be exploited to compromise the database.",
"raw": {
    "check_id": "python.flask.security.injection.tainted-sql-string.tainted-sql-string",
    "path": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sign",
    "start": {
        "line": 171,
        "col": 21,
        "offset": 7075
    },
    "end": {
        "line": 171,
        "col": 64,
        "offset": 7118
    },
    "extra": {
        "message": "Detected user input used to manually construct a SQL string. This is usually bad practice",
        "metadata": {
            "cwe": [
                "CWE-704: Incorrect Type Conversion or Cast"
            ],
            "owasp": [
                "A01:2017 - Injection",
                "A03:2021 - Injection",
                "A05:2025 - Injection"
            ],
            "references": [
                "https://docs.sqlalchemy.org/en/14/core/tutorial.html#using-textual-sql",
                "https://www.tutorialspoint.com/sqlalchemy/sqlalchemy_quick_guide.htm",
                "https://docs.sqlalchemy.org/en/14/core/tutorial.html#using-more-specific-text-with-table-expressions"
            ],
            "category": "security",
            "technology": [
                "sqlalchemy",
                "flask"
            ],
            "subcategory": [
                "vuln"
            ],
            "impact": "MEDIUM",
            "likelihood": "MEDIUM",
            "confidence": "MEDIUM",
            "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
            "vulnerability_class": [
                "Improper Validation"
            ],
            "source": "https://semgrep.dev/r/python.flask.security.injection.tainted-sql-string.tainted-sql-string",
            "shortlink": "https://sg.run/JxZj"
        },
        "severity": "ERROR",

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        "fingerprint": "requires login",
        "lines": "requires login",
        "validation_state": "NO_VALIDATOR",
        "engine_kind": "OSS"
    }
}
},
{
    "id": "17175b96-a84b-4793-b582-4f170fe59254",
    "title": "Xss in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sign",
    "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sign",
    "line_start": 163,
    "line_end": 163,
    "severity": "MEDIUM",
    "vuln_type": "XSS",
    "description": "Detected user input flowing into a manually constructed HTML string. You may be accident",
    "rule_id": "python.django.security.injection.raw-html-format.raw-html-format",
    "confidence": 0.7,
    "cwe_id": "CWE-79: Improper Neutralization of Input During Web Page Generation ('Cross-site Scripting')",
    "cvss_score": null,
    "code_snippet": "        161 |         content = \"<h2>User Lookup</h2>\"\\n        162 |         content += '<form act",
    "sources": [
        "semgrep"
    ],
    "risk_score": 0.3478,
    "risk_category": "LOW",
    "category_source": "score",
    "category_reason": "score-threshold",
    "llm_explanation": "Cross-Site Scripting (XSS) allows attackers to inject malicious scripts into web pag",
    "fix_suggestion": "",
    "business_impact": "An attacker could steal session tokens from user accounts and personal data, redirec",
    "raw": {
        "check_id": "python.django.security.injection.raw-html-format.raw-html-format",
        "path": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-s",
        "start": {
            "line": 163,
            "col": 16,
            "offset": 6750
        },
        "end": {
            "line": 163,
            "col": 87,
            "offset": 6821
        }
    },
    "extra": {
        "message": "Detected user input flowing into a manually constructed HTML string. You may be accident",
        "metadata": {
            "cwe": [
                "CWE-79: Improper Neutralization of Input During Web Page Generation ('Cross-site Scripting')"
            ],
            "owasp": [
                "A07:2017 - Cross-Site Scripting (XSS)",
                "A03:2021 - Injection",
                "A05:2025 - Injection"
            ],
            "category": "security",
            "technology": [
                "django"
            ],
            "references": [
                "https://docs.djangoproject.com/en/3.2/topics/http/shortcuts/#render",
                "https://docs.djangoproject.com/en/3.2/topics/security/#cross-site-scripting-xss-protection"
            ],
            "cwe2022-top25": true,
            "cwe2021-top25": true,
            "subcategory": [
                "vuln"
            ],
            "likelihood": "HIGH",
            "impact": "MEDIUM",
            "confidence": "MEDIUM",

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        "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
        "vulnerability_class": [
            "Cross-Site-Scripting (XSS)"
        ],
        "source": "https://semgrep.dev/r/python.django.security.injection.raw-html-format.raw-html-format",
        "shortlink": "https://sg.run/oYj1"
    },
    "severity": "WARNING",
    "fingerprint": "requires login",
    "lines": "requires login",
    "validation_state": "NO_VALIDATOR",
    "engine_kind": "OSS"
}
}
},
{
    "id": "3e68861b-6112-49a9-9817-1d957d50fac7",
    "title": "Unknown in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\",
    "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sign",
    "line_start": 124,
    "line_end": 143,
    "severity": "MEDIUM",
    "vuln_type": "UNKNOWN",
    "description": "Found a template created with string formatting. This is susceptible to server-side temp",
    "rule_id": "python.flask.security.audit.render-template-string.render-template-string",
    "confidence": 0.7,
    "cwe_id": "CWE-96: Improper Neutralization of Directives in Statically Saved Code ('Static Code Injection)",
    "cvss_score": null,
    "code_snippet": "        122 |          # \u274c Missing security headers (X-Frame-Options, CSP, etc.)\n        123 |",
    "sources": [
        "semgrep"
    ],
    "risk_score": 0.3384,
    "risk_category": "LOW",
    "category_source": "score",
    "category_reason": "score-threshold",
    "llm_explanation": "A security vulnerability was detected in your code. Found in vulnerable_server.py. R",
    "fix_suggestion": "",
    "business_impact": "This vulnerability could be exploited to compromise user accounts and personal data",
    "raw": {
        "check_id": "python.flask.security.audit.render-template-string.render-template-string",
        "path": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-s",
        "start": {
            "line": 124,
            "col": 26,
            "offset": 4583
        },
        "end": {
            "line": 143,
            "col": 9,
            "offset": 5708
        },
        "extra": {
            "message": "Found a template created with string formatting. This is susceptible to server-side temp",
            "metadata": {
                "cwe": [
                    "CWE-96: Improper Neutralization of Directives in Statically Saved Code ('Static Code Injection)",
                ],
                "owasp": [
                    "A03:2021 - Injection",
                    "A05:2025 - Injection"
                ],
                "references": [
                    "https://nvisium.com/blog/2016/03/09/exploring-ssti-in-flask-jinja2.html"
                ],
                "category": "security",
                "technology": [
                    "flask"
                ],
                "subcategory": [
                    "audit"
                ]
            }
        }
    }
}

```

```

    ],
    "likelihood": "LOW",
    "impact": "MEDIUM",
    "confidence": "LOW",
    "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
    "vulnerability_class": [
        "Code Injection"
    ],
    "source": "https://semgrep.dev/r/python.flask.security.audit.render-template-string.render-templat",
    "shortlink": "https://sg.run/8yjE"
},
"severity": "WARNING",
"fingerprint": "requires login",
"lines": "requires login",
"validation_state": "NO_VALIDATOR",
"engine_kind": "OSS"
}
}
},
{
    "id": "604172d0-d417-4997-b35c-02e26f9c6295",
    "title": "Unknown in C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\",
    "file": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-sign",
    "line_start": 321,
    "line_end": 321,
    "severity": "MEDIUM",
    "vuln_type": "UNKNOWN",
    "description": "Detected a dynamic value being used with urllib. urllib supports 'file://' schemes, so a",
    "rule_id": "python.lang.security.audit.dynamic-urllib-use-detected.dynamic-urllib-use-detected",
    "confidence": 0.7,
    "cwe_id": "CWE-939: Improper Authorization in Handler for Custom URL Scheme",
    "cvss_score": null,
    "code_snippet": "        319 |             try:\\n        320 |                 # \\u274c VULNERABLE: No URL validation",
    "sources": [
        "semgrep"
    ],
    "risk_score": 0.3384,
    "risk_category": "LOW",
    "category_source": "score",
    "category_reason": "score-threshold",
    "llm_explanation": "A security vulnerability was detected in your code. Found in vulnerable_server.py. D",
    "fix_suggestion": "",
    "business_impact": "This vulnerability could be exploited to compromise user accounts and personal data",
    "raw": {
        "check_id": "python.lang.security.audit.dynamic-urllib-use-detected.dynamic-urllib-use-detected",
        "path": "C:\\Users\\Dhruv\\Desktop\\College Stuff\\Projects\\CyberScurity Project - Copy\\security-s",
        "start": {
            "line": 321,
            "col": 20,
            "offset": 13906
        },
        "end": {
            "line": 321,
            "col": 58,
            "offset": 13944
        },
        "extra": {
            "message": "Detected a dynamic value being used with urllib. urllib supports 'file://' schemes, so a",
            "metadata": {
                "cwe": [
                    "CWE-939: Improper Authorization in Handler for Custom URL Scheme"
                ],
                "owasp": "A01:2017 - Injection",
                "source-rule-url": "https://github.com/PyCQA/bandit/blob/b1411bfb43795d3fffd268bef17a839dee954c2b1",
                "bandit-code": "B310",
                "asvs": {
                    "control_id": "5.2.4 Dynamic Code Execution Features",
                    "control_url": "https://github.com/OWASP/ASVS/blob/master/4.0/en/0x13-V5-Validation-Sanitization",
                    "section": "V5: Validation, Sanitization and Encoding Verification Requirements",
                    "version": "4"
                }
            }
        }
    },

```

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    "category": "security",
    "technology": [
      "python"
    ],
    "references": [
      "https://cwe.mitre.org/data/definitions/939.html"
    ],
    "subcategory": [
      "audit"
    ],
    "likelihood": "LOW",
    "impact": "LOW",
    "confidence": "LOW",
    "license": "Semgrep Rules License v1.0. For more details, visit semgrep.dev/legal/rules-license",
    "vulnerability_class": [
      "Improper Authorization"
    ],
    "source": "https://semgrep.dev/r/python.lang.security.audit.dynamic-urllib-use-detected.dynamic-u",
    "shortlink": "https://sg.run/dKZZ"
  },
  "severity": "WARNING",
  "fingerprint": "requires login",
  "lines": "requires login",
  "validation_state": "NO_VALIDATOR",
  "engine_kind": "OSS"
}
}
}
```